

VOLTAGE STABILITY EVALUATION USING MODAL ANALYSIS

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Abstract - This paper discusses the voltage stability analysis of large power systems using a modal analysis technique. The method computes, using a steady state system model, a specified number of the smallest eigenvalues and the associated eigenvectors of a reduced Jacobian matrix. The eigenvalues, each of which is associated with a mode of voltage/reactive power variation, provide a relative measure of proximity to voltage instability. The eigenvectors are used to describe the mode shape and to provide information about the network elements and generators which participate in each mode. A simultaneous iteration method, which is well suited to applications involving large power systems, is used for selective calculation of appropriate eigenvalues. Results obtained using a 3700 bus test system are presented illustrating the applicability of the approach.

Key Words - Reactive Power - Voltage Stability - Eigenvalues - Eigenvectors - Modal Analysis - Modelling - Large Systems

1. INTRODUCTION

As power systems are operated under increasingly stressed conditions, the ability to maintain voltage stability becomes a growing concern. In planning and operating power systems, the analysis of voltage stability for a given system state involves the examination of two aspects;

- a) *Proximity* : how close is the system to voltage instability ?
- b) *Mechanism* : when voltage instability occurs, what are the key contributing factors, what are the voltage-weak points, and what areas are involved ?

Proximity gives a measure of voltage security whereas *mechanism* provides information useful in determining system modifications or operating strategies which could be used to prevent voltage instability.

Voltage stability is indeed a dynamic phenomenon and can be studied using extended transient/midterm stability simulations. However, such simulations do not readily provide sensitivity information or the degree of stability. They are also time consuming in terms of CPU and engineering required for analysis of results. Therefore, the application of dynamic simulations is limited to investigation of specific voltage collapse situations, including fast or transient voltage collapse, and for coordination of protection and controls.

Voltage stability analysis often requires examination of a wide range of system conditions and a large number of contingency scenarios. For such applications, the approach based on

steady state analysis is more attractive and, if used properly, can provide much insight into the voltage/reactive power problem.

A number of special algorithms have been proposed in the literature for voltage stability analysis using the static approach [1-5]. In general, these have not found widespread practical application, and utilities tend to depend largely on conventional powerflow programs to determine voltage collapse levels of various points in a network. However, this approach is laborious and does not provide sensitivity information useful in making design decisions.

Some utilities use Q-V curves at a small number of load buses to determine the proximity to voltage collapse and to establish system design criteria based on Q and V margins determined from the curves [6]. One problem with the Q-V curve method is that it is generally not known apriori at which buses the curves should be generated. In producing a Q-V curve, the system in the neighborhood of the bus is unduly stressed and results may be misleading. In addition, by focussing on a small number of buses, system-wide problems may not be readily recognized.

An approach using V-Q sensitivity and piecewise linear powerflow analysis to find the margin, measured in terms of total load growth, between a given operating condition and the voltage collapse point is described in [7]. There has been some indication that the linearized powerflow solution may not be sufficiently accurate as the collapse point is approached. Also, V-Q sensitivity information, as shown in this paper, could be misleading when applied to a large system having more than one area with voltage stability problems.

Most of the approaches proposed to date use conventional powerflow models to represent the system steady state. This may not always be appropriate, especially as the system approaches critical condition. There is a need to consider more detailed steady state models for key system components such as generators, SVCs, induction motors and voltage dependent static loads. Load characteristics in particular could be critical and expanded subtransmission representation in the voltage collapse areas may be necessary [8].

As noted in [9], there is a need for analytical tools capable of predicting voltage collapse in complex networks, accurately quantifying stability margins and power transfer limits, identifying voltage-weak points and areas susceptible to voltage instability, and identifying key contributing factors and sensitivities that provide insight into system characteristics to assist in developing remedial actions.

This paper describes a modal analysis approach with the objective of meeting the above requirements. It involves the computation of a small number of eigenvalues and the associated eigenvectors of a reduced Jacobian matrix which retains the Q-V relationships in the network and which includes the appropriate characteristics of generators, loads, reactive power compensating devices, and HVDC converters. This parallels the use of modal analysis for small signal stability studies [10]. However, by using the reduced

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Jacobian instead of the system state matrix, the focus is on voltage and reactive power characteristics. The eigenvalues of the Jacobian identify different modes through which the system could become voltage unstable. The magnitude of the eigenvalues provide a relative measure of *proximity* to instability. The eigenvectors, on the other hand, provide information related to the *mechanism* of loss of voltage stability. Fast analytical algorithms for selective computation of a specified number of the smallest eigenvalues make the approach suitable for the analysis of large complex power systems. In addition, by appropriate choice of system models, the method can be used to analyze *snapshot* powerflows representing different time frames following major system disturbances.

2. MODAL ANALYSIS FOR VOLTAGE STABILITY EVALUATION

A system is voltage stable at a given operating condition if for every bus in the system, bus voltage magnitude increases as reactive power injection at the same bus is increased. A system is voltage unstable if, for at least one bus in the system, bus voltage magnitude decreases as the reactive power injection at the same bus is increased. In other words, a system is voltage stable if V-Q sensitivity is positive for every bus and unstable if V-Q sensitivity is negative for at least one bus.

Reduced Jacobian Matrix

The linearized steady state system power voltage equations are given by,

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J_{P\theta} & J_{PV} \\ J_{Q\theta} & J_{QV} \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix} \quad (1)$$

where,

ΔP = incremental change in bus real power.

ΔQ = incremental change in bus reactive power injection.

$\Delta \theta$ = incremental change in bus voltage angle.

ΔV = incremental change in bus voltage magnitude.

If the conventional powerflow model is used for voltage stability analysis, the Jacobian matrix in (1) is the same as the Jacobian matrix used when the powerflow equations are solved using the Newton-Raphson technique. With enhanced device models included, the elements of the Jacobian matrix in (1) are modified as discussed in section 3.

System voltage stability is affected by both P and Q. However, at each operating point we keep P constant and evaluate voltage stability by considering the incremental relationship between Q and V. This is analogous to the Q-V curve approach. Although incremental changes in P are neglected in the formulation, the effects of changes in system load or power transfer levels are taken into account by studying the incremental relationship between Q and V at different operating conditions.

To reduce (1), let $\Delta P=0$, then,

$$\begin{aligned} \Delta Q &= [J_{QV} - J_{Q\theta} J_{P\theta}^{-1} J_{PV}] \Delta V \\ &= J_R \Delta V \end{aligned} \quad (2)$$

and,

$$\Delta V = J_R^{-1} \Delta Q \quad (3)$$

where,

$$J_R = [J_{QV} - J_{Q\theta} J_{P\theta}^{-1} J_{PV}] \quad (4)$$

J_R is called the reduced Jacobian matrix of the system. J_R is the matrix which directly relates the bus voltage magnitude and bus reactive power injection. Eliminating the real power

and angle part from the system steady state equations allows us to focus on the study of the reactive demand and supply problem of the system as well as minimize computational effort.

The program developed also provides the option of performing eigen-analysis of the full Jacobian matrix. If the full Jacobian is used, however, the results represent the relationship between $(\Delta \theta, \Delta V)$ and $(\Delta P, \Delta Q)$. Since $\Delta \theta$ is included in the formulation, it is difficult to discern the relationship between ΔV and $(\Delta P, \Delta Q)$ which is of primary importance for voltage stability analysis. Also modal analysis using the full Jacobian matrix is computationally more expensive than using the reduced Jacobian. For these reasons we have chosen the reduced Jacobian approach.

Modes of Voltage Instability

Let,

$$J_R = \xi \wedge \eta \quad (5)$$

where,

ξ = right eigenvector matrix of J_R .

η = left eigenvector matrix of J_R .

$\wedge$ = diagonal eigenvalue matrix of J_R .

and,

$$J_R^{-1} = \xi \wedge^{-1} \eta \quad (6)$$

From (3) and (6), we have,

$$\Delta V = \xi \wedge^{-1} \eta \Delta Q \quad (7)$$

or,

$$\Delta V = \sum_i \frac{\xi_i \eta_i}{\lambda_i} \Delta Q \quad (8)$$

Where ξ_i is the i^{th} column right eigenvector and η_i the i^{th} row left eigenvector of J_R .

Similar to the concept used in linear dynamic system analysis [10], each eigenvalue λ_i , and the corresponding right and left eigenvectors ξ_i and η_i , define the i^{th} mode of the system. The i^{th} modal reactive power variation is,

$$\Delta Q_{m1} = K_1 \xi_1 \quad (9)$$

where,

$$K_1^2 \sum_j \xi_{j1}^2 = 1 \quad (10)$$

with ξ_{j1} the j^{th} element of ξ_1 .

The corresponding i^{th} modal voltage variation is,

$$\Delta V_{m1} = \frac{1}{\lambda_1} \Delta Q_{m1} \quad (11)$$

It is seen that, when the reactive power variation is along the direction of ξ_1 , the corresponding voltage variation is also along the same direction and the magnitude is amplified by a factor which is equal to the magnitude of the inverse of the i^{th} eigenvalue. In this sense, the magnitude of each eigenvalue λ_i determines the weakness of the corresponding modal voltage. The smaller the magnitude of λ_i , the weaker the corresponding modal voltage. If $|\lambda_i| = 0$, the i^{th} modal voltage will collapse because any change in that modal reactive power will cause infinite modal voltage variation.

In (8), let $\Delta Q = e_k$, where e_k has all its elements zero except the k^{th} one being 1. Then,

$$\Delta V = \sum_i \frac{\eta_{ik} \xi_i}{\lambda_i} \quad (12)$$

with η_{ik} the k^{th} element of η_i .

V-Q sensitivity at bus k,

$$\begin{aligned} \frac{\partial V_k}{\partial Q_k} &= \sum_i \frac{\xi_{ki} \eta_{ik}}{\lambda_i} \\ &= \sum_i \frac{P_{ki}}{\lambda_i} \end{aligned} \quad (13)$$

A system is voltage *stable* if the eigenvalues of the Jacobian are all positive. Those who are used to small signal stability analysis using eigenvalue techniques may find the requirement for the eigenvalues of the Jacobian to be positive for voltage stability a little confusing because in the study of small signal stability, an eigenvalue with positive real part indicates that the system is *unstable*. The relationship between system voltage stability and eigenvalues of the Jacobian J_R is best understood by relating the eigenvalues of J_R with the V-Q sensitivities, (which must be positive for stability), at each bus.

For practical purposes, J_R can be taken as a symmetric matrix and therefore, the eigenvalues of J_R are close to being purely real. If all the eigenvalues are positive, J_R is positive definite thus V-Q sensitivities are also positive indicating that the system is voltage stable. As the system is stressed, the eigenvalues of J_R become smaller until, at the critical point of system voltage stability, at least one of the eigenvalues of J_R becomes zero.

If some of the eigenvalues of J_R are negative, the system has passed the critical point of voltage stability because the eigenvalues of J_R change continuously from positive to zero to negative as the system is stressed.

While the magnitude of the eigenvalues can provide a relative measure of the *proximity* to instability, they do not provide an absolute measure because of the non-linearity of the problem. This is analogous to the damping factor in small signal stability analysis, which is indicative of the degree of damping but is not an absolute measure of stability margin. If a megawatt distance to voltage instability is required, the system is stressed incrementally until it becomes unstable and modal analysis applied at each operating point. The application of modal analysis is to help in determining how stable the system is, how much extra load or power transfer level should be added and, when the system reaches voltage stability critical point, to determine the voltage stability critical areas and to describe the *mechanism* of instability by identifying elements which participate in each mode.

Bus Participations

The participation factor of bus k to mode i is defined as,

$$P_{ki} = \xi_{ki} \eta_{ik} \quad (14)$$

From (13), P_{ki} indicates the contribution of the i^{th} eigenvalue to the V-Q sensitivity at bus k. The bigger the value of P_{ki} , the more λ_i contributes in determining V-Q sensitivity at bus k. For all the small eigenvalues, bus participation factors determine the areas close to voltage instability.

Branch and Generator Participations

When the change in reactive power injection is ΔQ_{mi} , the resulting voltage variation is ΔV_{mi} , and, the i^{th} modal angle variation is,

$$\Delta \theta_{mi} = -J_{P\theta}^{-1} J_{PV} \Delta V_{mi} \quad (15)$$

With ΔV and $\Delta \theta$ known, the linearized reactive loss variation across transmission branch lj, ΔQ_{lji} , and the linearized reactive power output variation at generator gk, ΔQ_{gki} can be calculated.

Let,

$$\Delta Q_{lmaxi} = \max_j (\Delta Q_{lji})$$

$$\Delta Q_{gmaxi} = \max_k (\Delta Q_{gki})$$

The participation factor of branch lj to mode i is defined as,

$$P_{lji} = \frac{\Delta Q_{lji}}{\Delta Q_{lmaxi}} \quad (16)$$

Branch participations thus indicate, for each mode, which branches consume the most reactive power for a given incremental change in reactive load. Branches with high P_{lji} are those which cause mode i to be weak. The branch participations thus provide valuable information regarding: (i) remedial actions in terms of transmission branch enhancement and redistributing the power flow to alleviate the loading on that branch, and (ii) criteria for contingency selection.

The participation factor of generator gk to mode i is defined as,

$$P_{gki} = \frac{\Delta Q_{gki}}{\Delta Q_{gmaxi}} \quad (17)$$

Generator participations indicate, for each mode, which generators supply the most reactive output in response to an incremental change in system reactive load. Generators with high P_{gki} are important in maintaining stability of mode i.

Calculation of Eigenvalues and Eigenvectors of J_R

It is impractical and unnecessary to calculate all the eigenvalues of a system with several thousand buses. An algorithm for calculating the minimum singular value and the corresponding left and right singular vectors, for both the full Jacobian and the reduced Jacobian, has been developed in [14]. The problem with using the minimum singular value or the minimum eigenvalue as voltage stability index lies in the fact that for a large complex system there are usually more than one weak mode associated with different parts of the system. As a system is stressed, the mode associated with the minimum singular value or the minimum eigenvalue of the base case system may no longer be the most troublesome mode. If the m smallest eigenvalues of J_R are determined, we have obtained the m least stable modes of the system. If the biggest of the m eigenvalues, say mode m, is deemed a strong enough mode, the modes which are not calculated can be neglected because they are known to be stronger than mode m.

An Implicit Inverse Lop-sided Simultaneous Iteration (IILSI) technique is used to calculate the m smallest eigenvalues of J_R and associated right and left eigenvectors. The IILSI algorithm can be viewed as a combination of the simultaneous iteration method and implicit inverse iteration method.

Lop-sided simultaneous iteration method for calculating m eigenvalues with the largest magnitudes and the associated right eigenvectors for a general real matrix A can be summarized as follows [11],

- Select m initial trial vectors $R = [R_1, R_2, \dots, R_m]$.
- Premultiply R by A, $S = AR$.
- Determine $G = R^H R$, $H = R^H S$.
- Solve $GB = H$ for B.

- e. Do full eigen-solution of B.
- f. Determine $W = ST$, with T the right eigenvector matrix of B.
- g. Set $R = W^*$, where W^* is W normalized such that all the vectors have their largest element equal to unity.
- h. Check convergence by comparing the last two solutions of R. If converged, stop. Otherwise go back to b.

Upon convergence, the eigenvalues of B give the m largest eigenvalues of A, and R contains the corresponding right eigenvectors. The same procedure applied to A^T provides the m largest eigenvalues of A and the associated left eigenvectors.

Since we are interested in the smallest eigenvalues of J_R , which correspond to the largest eigenvalues of J_R^{-1} , the simultaneous iteration algorithm has to be applied to J_R^{-1} . At each iteration, the premultiplication is,

$$S = J_R^{-1} R \quad (18)$$

Recall $J_R = J_{QV} - J_{Q\theta} J_{P\theta}^{-1} J_{PV}$, which is not sparse due to the reduction. To fully exploit the sparsity of the Jacobian matrix, S in (18) is obtained by solving the following sparse linear equations,

$$\begin{bmatrix} J_{P\theta} & J_{PV} \\ J_{Q\theta} & J_{QV} \end{bmatrix} \begin{bmatrix} Z \\ S \end{bmatrix} = \begin{bmatrix} 0 \\ R \end{bmatrix} \quad (19)$$

The ILSI method is applied to J_R and $[J_R]^T$ in turn to calculate the smallest eigenvalues and the corresponding right and left eigenvectors. An alternative approach is to solve for the right and left eigenvectors simultaneously, which requires in each iteration loop the solutions of $J_R S = R$ and $[J_R]^T S' = R'$. Because J_R is very close to being symmetric, the iteration for left eigenvectors converge very fast starting with the right eigenvectors as trial vectors. Also, there may exist cases where only the right eigenvectors are of interest. We believe, therefore, that the lop-sided approach is more efficient and flexible than iterating on the right and left eigenvectors simultaneously.

3. DEVICE MODELLING

In this section, we briefly describe the models for key system devices which, for voltage stability analysis, differ from conventional powerflow models. Since our basic modal analysis approach is based on linearized steady state system reactive power voltage relationship, the focus will be on the modification of the system Jacobian matrix to include these new models.

The linear relationship between power and voltage for each device is given by,

$$\begin{bmatrix} \Delta P_d \\ \Delta Q_d \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \Delta V_d \\ \Delta \theta_d \end{bmatrix} \quad (20)$$

where,

ΔP_d = incremental change in device real power output.
 ΔQ_d = incremental change in device reactive power output.
 ΔV_d = incremental change in device voltage magnitude.
 $\Delta \theta_d$ = incremental change in device voltage angle.

The self terms of the network Jacobian matrix associated with each device are modified by A_{11} , A_{12} , A_{21} and A_{22} to form the system Jacobian matrix.

Generator Model

The synchronous machine model accounts for the field current limit, armature current limit and under excitation limit. When either the field current or armature current limit is reached, the relationship between ΔP , ΔQ and $\Delta \theta$, ΔV are

given by the steady state synchronous machine equations [12]. The under excitation limit is modelled as a constant lower reactive limit.

When a generator is operating normally without hitting any of the limits, the model accounts for the droop effect of the AVR. Provision is also made for representing load (or line drop) compensation [13].

For most studies, generator active power outputs are assumed constant. In studying contingencies which result in loss of load or generation however, the active power is adjusted using a governor response powerflow algorithm which accounts for frequency variation and governor characteristics of each generator.

Load Model

Voltage dependency of load real and reactive power may have large impact on system voltage stability [8]. The load is modelled as the sum of several components, each of which is a different exponential function of voltage:

$$P = \sum_i P_{oi} V^{\alpha_i} \quad (21)$$

$$Q = \sum_i Q_{oi} V^{\beta_i} \quad (22)$$

Induction Motor Model

An induction motor is modelled using its steady state equivalent circuit. The following relationship between the load torque(T) and slip(s) is assumed,

$$T_L = T_o (1-s)^m \quad (23)$$

The linearized relationship between P and Q and voltage at a given operating condition can be easily derived from the machine equivalent circuit and load torque/speed characteristic. If the operating condition is such that the machine stalls, $s=1$ and the motor becomes a constant impedance load.

SVC Model

There are three regions of operation with different steady state control characteristics responding to voltage change.

Linear control region : $Q = K(V_{REF} - V)$
 Constant capacitance region : $Q = Y_{max} V^2$
 Constant reactance region : $Q = -Y_{min} V^2$

The linearized Q-V characteristic in different regions are used to derive the corresponding diagonal elements of the Jacobian matrix.

Two Terminal HVDC Model

The steady state HVDC model used includes three types of converter characteristics, namely constant DC current with constant inverter voltage, constant extinction angle γ , and constant ignition angle α .

Snapshots of Events Following a Contingency

Voltage control devices such as ULTC and switchable shunt reactors and capacitors operate discontinuously, and the time delays associated with their operations are often long. The best way to study the effect of these devices, without going through detailed time domain simulation, is the so called snapshot approach. By appropriate choice of device models, system conditions at any instant can be approximated. Modal

analysis can be applied to study voltage stability of the system immediately following the contingency, after fast controls, after slow controls and after operator actions. The effects of ULTCs and switchable shunt devices on the system Jacobian matrix are taken into account in forming the Y matrix.

The snapshot approach also facilitates the consideration of different values of generator field current limits associated with different time frames following a contingency. For contingencies resulting in loss of generation or load, it allows consideration of governor response powerflows.

4. APPLICATIONS

The modal analysis method has been successfully applied to several large systems. For the 3700 bus case discussed here, the system was stressed by scaling up load in a specified region, and by redispatching real generation through a pre-defined loading schedule. Load was increased until the powerflow failed to converge. At the highest load level for which a powerflow solution was obtained (critical case), P-V curves and Q-V curves were examined to verify that the critical case was indeed on the verge of instability. Full modal analysis was conducted on the base case and the critical case. The ten smallest eigenvalues and associated right and left eigenvectors were obtained for each case. In addition, bus, branch and generator participations were calculated using the eigenvectors. All the results, unless otherwise specified, were obtained using steady state system model the same as the conventional powerflow model. CPU time required to compute one eigenvalue for a system this size is less than 60 seconds, and the time required to compute all ten eigenvalues and associated participations is less than 20 minutes (on VAX8650). Modal analysis has been performed on three large complex networks, representing different interconnected systems. A wide variety of system conditions with different intertie power transfer levels and contingencies have been studied. The number of guard vectors used to calculate 10 eigenvalues was 4, and divergence of the simultaneous iteration algorithm due to several eigenvalues being close to each other has never been experienced. This indicates the reliability and robustness of the eigen-solution algorithm.

For each of the eigenvalues computed, the associated bus participations were examined to determine the size and location of the mode. Each mode was given a descriptive name for identification purposes, based on the geographic location of the mode. Table I provides results for the base case and the critical case.

Table I The Ten Weakest Modes for the Base Case and Critical Case

Base Case			Critical Case		
λ	Area	Mode	λ	Area	Mode
0.249	North	A	0.012	Central	D
0.252	East	B	0.249	North	A
0.279	East	C	0.252	East	B
0.315	Central	D	0.279	East	C
0.348	Southeast	E	0.348	Southeast	E
0.514	Southeast	F	0.485	Central	K
0.517	South	G	0.514	Southeast	F
0.609	Northwest	H	0.517	South	G
0.621	South	I	0.609	Northwest	H
0.642	South	J	0.620	South	I

It is seen from Table I that nine of the ten modes occurring in the base case are also present in the critical case. As the system is loaded however, the ranking of the modes changes as certain areas become more stressed. In this example, the system was loaded in the central area. It is clear that the central mode is the weakest mode for the critical case, since the eigenvalue associated with this mode is the smallest.

Bus Participations

Table II lists, for the base case, the five largest bus participations for each of the five weakest modes. Table III provides the same information for the critical case. Some modes, such as the north mode A, are very localized, containing a small number of buses with dominant participations and may occur due to load buses at the end of long radial transmission lines. If such localized modes are outside the region of load increase and do not contain reactive sources, they may be of little significance in the stability analysis. Other modes, such as the central mode D, are not as localized and contain a large number of buses with similar participations. The instability of such a non-localized mode involves a significant portion of the system. If a mode contains a large number of buses, the participation for each bus is generally small because the sum of all the bus participations is equal to unity. The size of a bus participation in a given mode indicates the effectiveness of remedial action, applied at that bus, in stabilizing the mode. The results also show that as the system is stressed, the bus participations for each mode do not change significantly, indicating that the shape of each mode is relatively independent of loading condition.

Table II Base Case Bus Participations

North Mode A $\lambda=0.249$		East Mode B $\lambda=0.252$		East Mode C $\lambda=0.279$		Central Mode D $\lambda=0.315$		Southeast Mode E $\lambda=0.348$	
Bus	P_{ki}	Bus	P_{ki}	Bus	P_{ki}	Bus	P_{ki}	Bus	P_{ki}
10590	0.244	1898	0.202	1882	0.700	12092	0.011	1031	0.299
10591	0.236	1819	0.185	1821	0.066	12095	0.011	1032	0.297
10592	0.196	1855	0.182	1841	0.060	12434	0.010	1030	0.225
10593	0.177	1820	0.174	1833	0.046	12435	0.010	1029	0.165
10606	0.126	1818	0.138	1822	0.030	12228	0.010	740	0.001

Table III Critical Case Bus Participations

Central Mode D $\lambda=0.012$		North Mode A $\lambda=0.249$		East Mode B $\lambda=0.252$		East Mode C $\lambda=0.279$		Southeast Mode E $\lambda=0.348$	
Bus	P_{ki}	Bus	P_{ki}	Bus	P_{ki}	Bus	P_{ki}	Bus	P_{ki}
12434	0.010	10590	0.244	1898	0.202	1882	0.700	1031	0.299
12435	0.009	10591	0.236	1819	0.185	1821	0.066	1032	0.297
12289	0.008	10592	0.196	1855	0.182	1841	0.060	1030	0.225
12290	0.008	10593	0.177	1820	0.174	1833	0.046	1029	0.165
12291	0.008	10606	0.126	1818	0.138	1822	0.030	740	0.001

Branch Participations

Tables IV and V give the ten branches with the largest participations for the central mode D in the base case and critical case respectively. For the base case, six of the ten branches are elements connecting large sources of generation to the 500 kV backbone of the transmission system. In the critical case, of the ten branches the top nine are all elements, including some transformers, connecting large sources of generation to the central system which surrounds the area associated with the central mode.

Table IV Base Case Branch Participations for Mode D, $\lambda = 0.315$

From Bus	From KV	To Bus	To KV	CCT	P_{jli}
10001	500	10008	500	1	1.000
10008	500	10014	500	1	0.837
10186	500	10184	22	T3	0.685
10017	500	10762	22	T4	0.685
10009	500	10750	500	1	0.456
10150	220	10202	24	1	0.379
10004	500	10005	500	1	0.369
10149	220	10200	24	T1	0.348
10003	500	10095	220	14	0.324
10003	500	10095	220	15	0.323

Table V Critical Case Branch Participations for Mode D, $\lambda = 0.012$

From Bus	From KV	To Bus	To KV	CCT	P_{jli}
10001	500	10008	500	1	1.000
10008	500	10014	500	1	0.843
10007	500	10015	500	1	0.558
10007	500	10015	500	2	0.554
10148	220	10201	24	1	0.499
10148	220	10199	24	1	0.499
10009	500	10750	500	1	0.486
10151	220	10204	24	1	0.465
10151	220	10205	24	1	0.461
10003	500	10095	220	1	0.337

Branch participations indicate which elements are critical to the stability of a given mode. To demonstrate this for the base case, contingency cases were examined in which branches with large participations were outaged and modal analysis performed. The branches selected were the four non-transformer branches with the highest participations in mode D listed in Table IV. Transformer branches were not tested because outage of these elements would result in loss of generation. Table VI shows that in all outage cases, the eigenvalue associated with mode D is reduced, or even becomes negative, indicating that the loss of the given element causes the central mode to approach or reach instability. It is also seen that eigenvalues associated with the modes in which these branches showed little or no participation, were essentially unaffected by the outages.

Table VI Effect of Outage of Lines with High Participations in Mode D

Mode	Base Case	λ with Line Outage			
	λ	Circuit A	Circuit B	Circuit C	Circuit D
A	0.249	0.249	0.249	0.249	0.249
B	0.252	0.252	0.252	0.252	0.252
C	0.279	0.279	0.279	0.279	0.279
D	0.315	-0.007	0.084	-0.021	0.243
E	0.348	0.348	0.348	0.348	0.348

Circuit A=10008 x 10001 cct 1 Circuit B=10008 x 10014 cct 1
Circuit C=10009 x 10750 cct 1 Circuit D=10004 x 10005 cct 1

Generator Participations

Tables VII and VIII give the ten generators with the largest participations for the central mode D in the base case and the critical case, respectively. Of the ten generators listed for the base case, the top eight are in the immediate area surrounding the central mode D, and the remaining two generators are connected to major transmission feeding this area. In the critical case, six of the ten top participating generators in the base case reach their reactive limits and therefore no longer participate. As a result, the other four generators listed in Table VII become the top participating generators in Table VIII. The other six participating generators in the critical case are connected to major transmission feeding the central area.

Table VII Base Case Generator Participations for Mode D, $\lambda = 0.315$

Bus	Unit	PG MW	QG MVAR	P_{gk1}
10726	1	946	533	1.000
10184	1	946	533	1.000
10202	1	540	199	0.812
10200	1	542	182	0.800
10204	1	540	174	0.744
10205	1	540	173	0.744
10201	1	542	183	0.719
10199	1	542	183	0.711
10196	1	251	84	0.546
10191	1	251	70	0.451

Table VIII Critical Case Generator Participations for Mode D, $\lambda = 0.012$

Bus	Unit	PG MW	QG MVAR	P_{gk1}
10204	1	540	306	1.000
10205	1	540	305	0.995
10201	1	542	312	0.989
10199	1	542	312	0.981
11407	1	550	26	0.547
11408	1	550	26	0.538
10420	1	1256	489	0.500
10051	1	850	308	0.491
10054	1	940	312	0.444
10053	1	940	310	0.437

Generator participations indicate which machines are critical to the stability of a given mode. Generators with high participations in a given mode are those which exhibit the greatest change in reactive generation for a given load increase in the area associated with the mode. In general, it can be expected that these represent the machines which must maintain reactive reserves to ensure stability of the mode. To demonstrate this for the base case, the maximum reactive limits of machines with high participations in the central mode D, were set equal to the actual reactive output of the machines, thereby simulating the loss of reactive reserve on these generators. Table IX shows the results of modal analysis for three cases in which the reactive reserves were

removed for the machines with the top two, four, and ten participations.

Table IX Effect of Removing Generator Reactive Reserves

Mode	λ for Base Case	λ with top 2 Generators MVAR Limited	λ with top 4 Generators MVAR Limited	λ with top 10 Generators MVAR Limited
A	0.249	0.249	0.249	0.249
B	0.252	0.252	0.252	0.252
C	0.279	0.279	0.279	0.279
D	0.315	0.263	0.196	-0.054
E	0.348	0.348	0.348	0.348

It is seen that limiting the reactive reserves of the generators critical to the central mode D has a profound effect on the size of the eigenvalue associated with the mode. In fact, when the ten machines with the largest participations lose reactive reserve, the eigenvalue associated with mode D becomes negative, indicating the mode is unstable. It is therefore concluded that generator participations are important in establishing security criteria in terms of reactive reserves.

Effect of Device Modelling

Table X shows the ten smallest eigenvalues when different device models are used in forming the system Jacobian matrix. Model 1 is the same as the conventional powerflow model. For model 2, five generators in the central region which have $Q_g = Q_{gmax}$ are modelled as machines with constant field current. Model 3 has the powerflow generator models but all the reactive loads in the central region are represented as 25% constant MVAR and 75% constant impedance.

It is seen from Table X that the central mode D becomes much more stable when 75% of the reactive loads in the central region are modelled as constant impedance when compared with constant MVAR. Constant field current generator model also tends to stabilize the central mode D when compared with constant Q model. The results indicate that device modelling does have a significant impact on the voltage stability of the system and should not be overlooked.

Table X Effect of Generator and Load Model in Critical Case

Model 1		Model 2		Model 3		Model 1: Generator = PV or PQ Load = Constant MVA
Mode	λ	Mode	λ	Mode	λ	
D	0.012	D	0.018	D	0.213	Model 2: Generator = PV or Constant I_{fd} Load = Constant MVA
A	0.249	A	0.249	A	0.251	
B	0.252	B	0.252	B	0.252	
C	0.279	C	0.279	C	0.279	
E	0.348	E	0.348	E	0.348	
K	0.485	K	0.496	F	0.514	Model 3: Generator = PV or PQ P Load = Constant MW Q Load = 25% Constant MVAR 75% Constant Z
G	0.514	F	0.514	G	0.517	
F	0.517	G	0.517	H	0.620	
H	0.609	H	0.609	I	0.626	
I	0.620	I	0.620	J	0.642	

Powerflow Snapshots

Table XI shows the results of modal analysis of powerflow snapshots representing different timeframes following a contingency. The contingency selected was the outage of the branch with the highest participation in the central mode (10001 x 10008=circuit A in Table VI). For all tests, loads in the central region were represented as P=constant MW and Q=75% constant Z, 25% constant MVAR.

The results of Snapshot 1 show that, following a contingency, the stability of the central mode D is increased when generator reactive limits are not respected. As the limits are imposed however, as in Snapshot 2, the stability of the central mode decreases. Mode D is stable in this snapshot, but

unstable in Table VI, due to the difference in reactive load modelling. In Snapshot 3, ULTC actions are enabled and, in this example, act to reduce the stability of the central mode as the secondary voltages are raised by tap action and the voltage dependent load component increases. Although taps act to restore the load level to precontingency values, the load retains its voltage dependency characteristics. This accounts for the improved stability of mode D in snapshot 3 compared to results shown in Table VI. If shunt capacitors are present on the secondary side of some transformers in a given mode, tap action could, depending on the amount of compensation relative to the amount of load, lead to an improvement in the stability of the mode.

Table XI Modal Analysis at Different Snapshots

	Base Case	Snapshot 1	Snapshot 2	Snapshot 3
Contingency	No	Yes	Yes	Yes
Q_{gen} Limits	Yes	No	Yes	Yes
ULTCs Enabled	Yes	No	No	Yes
	Mode λ	Mode λ	Mode λ	Mode λ
	A 0.250	B 0.252	A 0.250	A 0.250
	B 0.252	C 0.279	B 0.252	B 0.252
	C 0.279	E 0.348	C 0.279	C 0.279
	E 0.348	D 0.479	D 0.310	D 0.307
	D 0.387	F 0.514	E 0.348	E 0.348
	F 0.514	G 0.518	F 0.514	F 0.514
	G 0.517	H 0.620	G 0.517	G 0.517
	H 0.621	I 0.625	H 0.618	H 0.618
	I 0.627	J 0.644	I 0.625	I 0.626
	J 0.642	L 0.663	J 0.644	J 0.644

Comparison with V-Q Sensitivities and L Indices

V-Q sensitivities were calculated for all the load buses in both the base case and the critical case. The ten buses with the highest V-Q sensitivities are listed in Table XII. It is seen that V-Q sensitivities do not provide a clear indication of weak voltage areas because buses associated with different modes are mixed in the ordering. It is also important to point out that the CPU time required to compute the sensitivities for all the load buses is more than the CPU time required to compute the ten smallest eigenvalues and all the associated participations.

Table XII V-Q Sensitivities for Base Case and Critical Case

Base Case					Critical Case				
No.	Bus	KV	Area	dV/dQ	No.	Bus	KV	Area	dV/dQ
1	1882	6.9	East	3.021	1	1882	6.9	East	3.021
2	4000	34.5	South	1.220	2	12092	2.4	Central	1.403
3	10602	6.6	Northwest	1.158	3	12095	2.4	Central	1.401
4	1303	115.0	Southeast	1.079	4	4000	34.5	South	1.222
5	10585	6.6	North	0.997	5	12434	13.8	Central	1.220
6	12092	2.4	Central	0.976	6	12435	115.0	Central	1.183
7	12095	2.4	Central	0.975	7	10602	6.6	Northwest	1.159
8	5388	11.5	South	0.821	8	12432	13.8	Central	1.091
9	1819	69.0	East	0.817	9	12291	13.8	Central	1.089
10	1855	69.0	East	0.802	10	12296	13.8	Central	1.083

L indices defined in [2] were also calculated for all the load buses. Table XIII list the ten buses with the largest L indices for both the base case and the critical case. In the base case, the L index does not identify all the voltage weak areas shown in modal analysis. In the critical case, both the L index method and the modal analysis method identify the central area to be the least stable area. The calculation of L indices for all the load buses requires much less CPU time than modal analysis and, therefore, the L index method is probably a useful screening tool. L indices however, do not provide any information regarding branch and generator participations and, more importantly, the modelling capability of the method

is limited to the conventional powerflow models. In addition, in calculating L indices, it is unclear as to how generators reaching their reactive limits should be accounted for.

Table XIII L Indices for Base Case and Critical Case

Base Case					Critical Case				
No.	Bus	KV	Area	L_i	No.	Bus	KV	Area	L_i
1	10676	118.0	West	0.618	1	12434	13.8	Central	0.864
2	10690	27.0	West	0.604	2	12435	13.8	Central	0.856
3	12434	13.8	Central	0.582	3	12289	13.8	Central	0.821
4	12435	13.8	Central	0.578	4	12290	13.8	Central	0.819
5	12289	13.8	Central	0.561	5	12291	13.8	Central	0.817
6	12290	13.8	Central	0.560	6	12296	13.8	Central	0.815
7	12291	13.8	Central	0.560	7	12432	13.8	Central	0.807
8	12296	13.8	Central	0.559	8	12433	13.8	Central	0.801
9	12432	13.8	Central	0.553	9	12242	13.8	Central	0.690
10	12433	13.8	Central	0.550	10	12243	13.8	Central	0.690

Application of Modal Analysis to On-line Stability Assessment

The applications described above are conducted as off-line studies in much the same way as transient stability simulations are performed. It is conceivable, however, that modal analysis could find use in on-line environments. To reduce CPU requirements, the method could be used with reduced system representations. Used in conjunction with on-line powerflows, the modal analysis technique could provide indications of system conditions with potential voltage stability problems, and signal the need for operator action. More work is required to fully investigate the potential of the modal analysis method for on-line stability assessment.

5. CONCLUSIONS

A voltage stability assessment technique for large power systems using modal analysis is presented. The method computes a specified number of the smallest eigenvalues of a reduced Jacobian matrix, and the associated bus, branch, and generator participations. Based on tests conducted using the method, the following conclusions are reached:

- Each eigenvalue corresponds to a mode of voltage/reactive power variation.
 - The modes corresponding to small eigenvalues represent the modes most prone to loss of stability.
 - The magnitude of each small eigenvalue provides a relative measure of proximity to loss of voltage stability for that mode.
- Bus, branch and generator participations provide useful information regarding the mechanism of loss of stability.
 - Bus participations indicate which buses are associated with each mode.
 - Branch participations show which branches are important in the stability of a given mode. This provides insight into possible remedial actions as well as contingencies which may result in loss of voltage stability.
 - Generator participations indicate which machines must retain reactive reserves to ensure stability of a given mode.
- The method includes detailed component models which may have significant impact on system voltage stability.
- Modal analysis of powerflow snapshots can be used to evaluate the stability of a system at various instants following system disturbances.
- Comparison with L index method and V-Q sensitivity method shows that the modal analysis approach has certain advantages over these methods.
- The method is well suited to voltage stability assessment of large complex systems.

6. REFERENCES

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Discussion

M. K. Pal (Public Service Electric and Gas Company, Newark, N.J.): This paper describes the voltage stability evaluation through a modal analysis using a reduced power flow Jacobian. Although the term "modal analysis" conjures a picture of dynamic system formulation and analysis, the approach is based on a steady-state analysis as has been pointed out at the beginning of the paper. The prospect of any kind of stability analysis by a static approach is always attractive. However, before a static analysis can be employed in the study of a dynamic phenomenon, care must be exercised in validating the approach. The paper is well written. Only after successive readings do some fundamental problems become apparent. In view of the possible wide acceptance of the methodology and widespread use of the computer program described in the paper, the following observations and comments would perhaps be in order.

The approach of the paper involves the computation of a small number of the smallest eigenvalues and the associated eigenvectors of the reduced power flow Jacobian. The magnitude of the eigenvalues will, presumably, provide a relative measure of proximity to voltage instability, on the assumption that the singularity of the Jacobian, when an eigenvalue becomes zero, signifies a voltage stability limit. Although the literature is replete with claims to the contrary, it has never been correctly demonstrated that singularity of the power flow Jacobian (or any other Jacobian) has anything to do with voltage stability. All these claims are based on flawed mathematics, as has been clearly demonstrated in [A]. In fact, it is not difficult to show that using the models generally used in these analyses, one would not encounter voltage instability. Singularity of the power flow Jacobian signifies a power limit as determined by network laws. It may not have anything to do with voltage stability.

Voltage instability, like any other instability, is a dynamic phenomenon. This is acknowledged by the authors in the introduction of the paper. A rigorous analysis, therefore, would require a dynamic system formulation and the application of appropriate dynamic criteria. System response following any kind of disturbance or system change is governed by system dynamics. It is true that in the normal operating region, for certain system changes, especially when the changes are slow and the load characteristic is favorable, system response can be predicted from a purely static consideration. However, carrying this over to abnormal operating regions can lead to misconceptions.

There are many misconceptions around in the field of voltage stability. All of these resulted from an attempt to predict system performance, which is basically dynamic, using a static system formulation, or, stated more precisely, using a static model of the element (the load) that is driving the system response. One such misconception is reflected in the definition of voltage stability as given in the paper. It defines a system as voltage unstable if the voltage magnitude at a bus decreases as reactive injection is increased at that bus, implying that the voltage would actually decrease if reactive support were added under certain operating conditions, such as operating in the lower portion of the power-voltage (P-V) curve. Consideration of the relevant dynamics of the load would, however, show that this cannot happen in normal operation. It is true that under certain control actions, such as under the control of an SVC, while operating in the lower portion of the P-V curve, a lowering of the voltage set-point would be accompanied by an eventual increase in the reactive injection from the SVC. In other words, when the voltage magnitude does decrease with increase in reactive injection, the operation is actually stable.

Load characteristics can be critical in the voltage stability problem, as has been recognized by the authors. Certain load characteristics can be accounted for in the steady-state analyses. We are not, however, comfortable with the authors' handling of two specific load types, viz. the induction motor and LTC controlling a static load. Consider the case of the induction motor. The motor is replaced by the steady-state equivalent circuit and then this is reflected in the system Jacobian. The details of the procedure are not given in the paper. The procedure apparently applies to the full range of motor slip - from zero to unity. It is indicated that at standstill, i.e. when $s=1.0$, the motor will be reduced to a pure impedance. At other values of slip, we can think of two possibilities. One is to follow the same procedure as at standstill, using the value of slip calculated at the given operating condition. The induction motor is thus, in effect, reduced to a constant impedance load. It can be easily verified that, while the critical eigenvalues of the conventional power flow Jacobian

(loads represented as constant P and Q) steadily decrease as the loading is increased, at least one becoming zero at the critical loading point, the opposite happens when the loads are modeled as constant impedance and the Jacobian is modified accordingly.

The other possibility is to treat s as a variable and derive the linearized load-voltage relationship by eliminating Δs . However, there can be a problem with this approach also. As an illustration, consider the system shown in Fig.1, where the motor load is represented by an approximate equivalent circuit. For simplicity, we will assume a constant load torque. (Note that equation (23) has a problem; according to this equation, the motor can never stall, no matter what.)

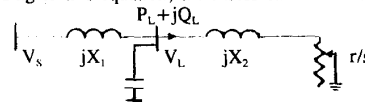


Fig.1 A simple two bus system supplying an induction motor load.

From Fig.1, $P_L + jQ_L = V_L[V_L/(r/s + jX_2)]^*$

Since the load torque is assumed constant, the real power drawn by the motor from the system is approximately constant in the steady state. Therefore, separating the real and imaginary parts, linearizing, and eliminating Δs , we obtain,

$$\Delta Q_L = k \Delta V_L \quad (1)$$

where

$$k = -\frac{2V_L X_2}{(r/s)^2 - X_2^2}$$

(1) can, therefore, be used to modify the power flow Jacobian. Note that k is negative for values of s in the range $0 < s < r/X_2$. In this range, the singularity of the modified Jacobian will correctly indicate voltage stability limit, assuming constant source voltage.

There is a problem, however, as s exceeds r/X_2 , when k changes sign. The relationship given by (1) then mimics SVC operation, aiding stability. The modified Jacobian under this condition will suggest stable operation, although the operation is clearly unstable.

In the simple example provided, the problem is easily detected. In large scale system studies using a computer and a more complex equivalent circuit for the induction motor, as done in the paper, this may escape detection, unless specific measures are taken.

In any case, the above procedure can give valid results only in small-signal stability studies and when the source voltage can be assumed to be constant. Actually, when induction motors comprise a significant part of the total load, and stability is in question, a detailed dynamic analysis including the effects of the rotor flux dynamics would be necessary. Also, induction motor loads respond fast; their speed of response is comparable to the speed of response of generator dynamics. Consequently, the assumption of constant generator terminal voltage is not valid. A detailed modeling of the generators and their excitation controls becomes necessary.

In voltage stability analyses of large systems, if the characteristics of the different types of loads are separately accounted for, then the generator dynamics must also be taken into account, since the effect of fast dynamics of part of the load is not being allowed to be canceled out by the stabilizing effect of the purely static loads.

As regards the LTC modeling, within its range, an LTC, controlling the load voltage, would tend to restore the voltage, thereby rendering a static, voltage dependent load, such as a constant impedance load, constant MVA. The effect is dynamic and would contribute to voltage collapse if the load exceeds the critical level, and the LTC limit is not reached. When the LTC is located at the load bus, and the load is constant MVA, LTC has no effect on voltage stability - viewed from the system side, tap changing has no effect. The inclusion of the LTC effect in the system Y matrix will have little impact on the eigenvalues. If the load is static, presumably, the voltage dependency would be accounted for in the way described in the paper, since the LTC effect is already included in the Y matrix. The dynamics of the loads would thus be completely ignored. The conclusion drawn from the analysis of the paper would, therefore, be exactly opposite of what would be experienced in actual operation.

There is also the question of large-disturbance stability which, in the case of voltage stability, may assume more importance than small-signal stability and which will not be addressed by the methodology of the paper. As it stands, the methodology described in the paper may, in some form, be suitable for use in the reactive power/voltage control problem, such as in determining the appropriate locations for installing additional reactive support. However, more powerful tools are already available for this purpose.

- [A] M. K. Pal, Discussion of "An Investigation of Voltage Instability Problems," by N. Yorino, H. Sasaki, Y. Masuda, Y. Tamura, M. Kitagawa and A. Oshimo, 91 WM 202-2 PWRs, IEEE/PES winter meeting February, 1991.

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Adam Semlyen (University of Toronto): The paper presents a powerful computational approach for the investigation of the voltage stability of a power system. It contains several interesting ideas and capabilities of which I have found the following as most remarkable:

— A classification of voltage stability problems to include, beside the aspect of *proximity*, also that of *mechanism*. The latter is served by computing additional information about the system in the form of eigenvectors and participation factors.

— The realization that more than a single eigenvalue and the corresponding eigenvectors are of significance in the assessment of the situation prior to reaching the immediate proximity of the voltage collapse point; and the production of the appropriate computational tools.

— The inclusion in the system Jacobian of effects representing the characteristics of terminal components (loads and generators).

— The reduction of the system Jacobian so that the real power related dynamics are effectively removed and the resultant reduced Jacobian J_R reflects only V - Q related effects.

— The interpretation given to J_R relating it to V - Q sensitivity effects rather than to V - Q dynamics.

In the following I would like to address the topic of the existence of a number of small eigenvalues before reaching a state close to voltage collapse and of the practical interest in their eigenspace (spanned by the corresponding eigenvectors). It is convenient for the examination of the related eigenvalue problem

$$Ae = 0 \quad (1)$$

to start with a matrix that has several zero eigenvalues as, for example,

$$A_{000} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 7 & 9 & 11 & 13 & 15 \\ 13 & 16 & 19 & 22 & 25 \\ 20 & 25 & 30 & 35 & 40 \end{bmatrix} \quad (2)$$

or, perhaps, with a similar, but symmetrical matrix.

It can be easily verified that A_{000} has three zero-eigenvalues ($\lambda_1 = 81.8552$, $\lambda_2 = -0.85517$, $\lambda_3 = \lambda_4 = \lambda_5 = 0$) and thus has rank 2. There is no unique solution for the eigenvectors e of equation (1), related to the zero eigenvalues. They are arbitrary but located in a three-dimensional subspace E_3 of $\mathbb{R}^5$. Therefore, e (normalized by setting its last element equal to unity) is a function of two arbitrary parameters, p_1 and p_2 , and contains only two unknowns, x_1 and x_2 :

$$e = [x_1 \ x_2 \ p_1 \ p_2 \ 1]^T = [z^T; \ p^T]^T \quad (3)$$

Its calculation will be shown in the sequel. Next, however, I will try to relate this problem to the eigenanalysis of the matrix J_R of a power system.

If we disturb A_{000} of equation (2) by adding a random matrix R multiplied by a small scalar ϵ ($\epsilon \ll 1$), all three zero-eigenvalues, λ_3 , λ_4 , and λ_5 , become non-zero (and random, but deterministically related to the random matrix R). Since they remain small, we denote the resultant matrix by A_{eee} . The small perturbation has only slightly modified the largest eigenvalues λ_1 and λ_2 , and the related eigenvectors, e_1 and e_2 . However, the eigenvectors e_3 , e_4 , and e_5 , corresponding to λ_3 , λ_4 , and λ_5 , will now be well defined by the associated eigenvalues. Nevertheless, they continue to span essentially the same subspace E_3 that represented the eigenspace of A_{000} for the three zero-eigenvalues; see page 307 of reference [A]. The conclusion, at this stage, is that what is essential and (almost) fixed in a situation that is not yet close to critical, is the eigenspace corresponding to the small eigenvalues rather than the individual eigenvectors themselves. Therefore, instead of calculating the eigenvectors of A_{eee} (or of J_R), it is sufficient to solve the much simpler eigenproblem (1) for the *parametrized* eigenvector of equation (3).

As the power system is loaded, J_R is subject to a one-parameter modification. By analogy, we may consider a continuous, one-parameter modification of A_{eee} . A particular such procedure is subtracting the same

quantity λ from each diagonal element. As $|\lambda|$ is gradually increased, criticality is eventually reached when the resulting matrix becomes singular. This, of course, corresponds to conventional eigenanalysis. However, loading of a power system does not translate into uniform diagonal changes of the Jacobian matrix J_R . This is why, in the studies presented in the paper, it was not the smallest eigenvalue that got closest to zero as the system was loaded. Clearly, *the smallest eigenvalue is not necessarily the best indicator of proximity to voltage collapse*. However, no matter what is the manner of continuous, one-parameter modification of the matrix, A_{eee} or J_R , criticality is reached by a single eigenvalue becoming zero. This is in contrast with the single, random perturbation that has simultaneously rendered non-zero all three zero-eigenvalues. If we would have perturbed only one element of matrix (2), only one zero-eigenvalue would have become non-zero; and perturbation of two elements (not in the same row and column) would have been needed to render non-zero two of the zero-eigenvalues.

At this stage, we see that the objectives of voltage stability analysis can be achieved by assessing the proximity to collapse and by finding the eigenspace E of the small eigenvalues. The eigenvalues and the eigenvectors themselves are then not needed.

The eigenspace E can be found by the following procedure. Rewrite equation (1), for the perturbed matrix A_{eee} and taking (3) into account, in the partitioned form

$$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \begin{bmatrix} x \\ p \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix} \quad (4)$$

Any selection of the rows of A_{eee} can in fact be used to form equation (4). Here r represents a residual vector since the matrix A_{eee} is not singular. The norm of r is thus a measure of the proximity to criticality, a Proximity Index, PI; see reference [B]. We solve problem (4) by first solving for x from the first part of (4), which we write in the form

$$A_{11}x = -A_{12}p \quad (5a)$$

This gives x as function of p . Its substitution into (3) gives e as a function of parameters p_i , the elements of p , for each of its elements. Substitution of x into the second equation of (4), gives

$$r = \text{lin}\{p\} \quad (5b)$$

i.e., the elements of r result as linear expressions of p_i .

In the case of the perturbed matrix

$$A_{eee} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 7 & 9 - \epsilon & 11 & 13 & 15 \\ 13 & 16 & 19 + 3\epsilon & 22 & 25 \\ 20 & 25 & 30 & 35 & 40 + \epsilon \end{bmatrix} \quad (6)$$

we obtain from (5a) (A_{11} being the first 2×2 submatrix of A_{eee})

$$x_1 = p_1 + 2p_2 + 3 \quad (7)$$

$$x_2 = -2p_1 - 3p_2 - 4$$

and then, from (3),

$$e = \begin{bmatrix} p_1 + 2p_2 + 3 \\ -2p_1 - 3p_2 - 4 \\ p_1 \\ p_2 \\ 1 \end{bmatrix} \quad (8a)$$

With this expression for e , the second equation of (4) gives, as in (5b), the residue vector

$$r = \epsilon \begin{bmatrix} 2p_1 + 3p_2 + 4 \\ 3p_1 \\ 1 \end{bmatrix} \quad (8b)$$

The Euclidian norm of the residue can be minimized by minimizing each element of the vector (8b). Here it means setting to zero the first two equations involved. This yields $p_1 = 0$ and $p_2 = -4/3$. With these values, (8a) gives the critical direction of e :

$$e_{cr} = [1 \ 0 \ 0 \ -4 \ 3]^T \quad (9)$$

The direction e_{cr} represents the particular eigenvector of the eigenspace E that corresponds to $PI = \min$, i.e., to minimal distance to voltage instability. In the general case, one may proceed to determine a (possibly complete) ordered sequence (with increasing PI) of orthogonal

eigenvectors e , with increasing PI, contained in E . The procedure described above can be generalized for any system, with the remark that the minimization of $\|r\|$ is in general a Linear Least Squares problem (with one more equation than unknowns) of small dimension and straightforward solution.

The eigenspace approach described above has the advantage that it requires essentially a single sparse matrix factorization and only a few backsubstitutions. It has thus the potential of solving in a computationally efficient way the eigenanalysis problem needed in voltage stability studies.

[A] G.W. Stewart, "Introduction to Matrix Computations", Accademic Press, 1973.

[B] A. Semlyen, "Imminent Voltage Instability with Large Induction Motor Loads", Proceedings of the International Workshop on Bulk Power System Voltage Phenomena: Voltage Stability and Security, held on August 4-7, 1991, at Deep Creek Lake, Maryland.

Manuscript received April 14, 1992.

C. Concordia (Consulting Engineer, 702 Bird Bay Drive West, Venice, Florida, 34292): This paper is an admirable example of attacking the problem of possible voltage instability of a large complex network in a practical and relatively simple and straightforward way. We have just two comments.

First, as a linearized model of the system, it must be treated with caution and the results interpreted in a system-specific way. In particular, as shown in Figs. 1.6 and 1.7 (pp. 13, 14) of the Working Group report (paper ref. 9), because of different methods of voltage support. "The margin cannot be deduced from the voltage behaviour" alone.

Further, since I think we should be interested in assessment of proximity as well as in limits, perhaps the definition should include some mention of danger even when dV/dQ is still positive but becoming very large. (This is mentioned in the discussion following eq. 13. Incidentally, although the paper distinguishes its method from the " $V-Q$ " sensitivity method, I should call it itself a $V-Q$ sensitivity method.)

Second, although I should hesitate to compete with the solution of a 3700-bus network, it seemed useful to compare the results for a 1-bus network as shown in Fig. 1.1 of the W.G. report.

It can be shown¹ that for a system with E (system internal voltage) and B (system short-circuit admittance) both equal to 1.0 and with load P and Q_C (reactive net power = support minus load) on the short-circuit power base, there is the very simple relation:

$$V^2 - Q_C = \sqrt{V^2 - P^2} \quad (1)$$

whence:

$$dV = \frac{\sqrt{V^2 - P^2} dQ_C - PdP}{[2\sqrt{V^2 - P^2} - 1]V} \quad (2)$$

Then for stability:

$$P^2 \leq V^2 - \frac{1}{4} \quad (3)$$

At the limit:

$$Q_C = V^2 - \frac{1}{2} = P^2 - \frac{1}{4} \quad (4)$$

A few things can be learned from this extremely simplified case. For example, if we take a point very near to the stability limit at $V = 1.0$. Then:

$$dV \approx \frac{0.5dQ_C - 0.866dP}{0 +}$$

We can see that:

- (1) The sensitivity dV/dQ_C and dV/dP are very roughly the same.
- (2) Controlled compensation (so that Q_C would increase as P increased) flattens the dV/dP relation so that it becomes less effective as a proximity indicator (in operation).

We have made these two comments, not to detract from the importance of having available practical means for studying large networks, but to remind ourselves of the concurrent need for a basic understanding of the nature of the problem.

Reference

- [1] "Voltage Stability Simplified"—a memorandum (unpublished) prepared a few years ago to clarify my own understanding and purporting to show "universal" $Q-V$, $P-V$, required Q , and dV/dP (dV/dQ) curves to serve as paradigms.

Manuscript received August 8, 1991.

Y. Mansour (B. C. Hydro, Vancouver, Canada): This paper is a welcome contribution on a subject that seems to be governing many utilities system plans and operation around the world. For years a general feeling that the characteristics of the load flow Jacobian, especially when modified to account for some slow dynamic effects, should contain enough information for voltage stability assessment and could provide important in-sight. Earlier efforts have led the way by calculating the minimum singular value or the minimum eigenvalue of the load flow Jacobian. Computation deficiency and modeling limitation limited the application. This paper made significant contributions to most of the relevant issues. At B. C. Hydro, we were fortunate to participate closely to the development reported by the authors and would like to share our experience to compliment the authors conclusions in some parts, support in some, and would appreciate the authors comments on others.

1. We share the authors experience that eigenvalues do not provide an absolute measure of proximity to instability. The behavior of the eigenvalues as the system is stressed is mostly nonlinear especially close to the limit.
2. We re-emphasize the authors conclusion that tracking the minimum eigenvalue or singular value could be misleading. We had the same experience with the B. C. Hydro's system where the most troublesome mode changes as the system is stressed. The idea of calculating the "m" least stable modes is better but the choice of m has been experimental and arbitrary. Even with 10 minimum eigenvalues calculation, we found in some cases that at the point of collapse a mode corresponding to the minimum eigenvalue which did not show earlier and suddenly takes control. It was also noticed that many of the minimum "m" eigenvalues are associated with localized problems related to buses located at the end of subtransmission feeders. Perhaps an educated search for minimum "m" eigenvalues associated with more global modes would focus on the relevant eigenvalues. This could be an important future development.
3. The value of modal analysis in identifying the areas vulnerable to voltage collapse and the candidate buses for most effective action has been proved: the participation vectors provided information for SVC application.
4. We have also used modal analysis in identifying the clusters of the system for area voltage control application with noticeable consistency. Earlier efforts reported in the literature have been tried without much success.
5. It is important to recognize the value of the extended modeling efforts, reported in the paper, beyond those used in traditional load flow studies. The load flow generator models as "PV" buses with minimum and maximum reactive power limits could be considerably misleading. Accounting for excitation and armature current limits as suggested by the authors is very important.
6. We have experienced cases where the load flow solution diverges while the minimum 10 eigenvalues are all positive. Could the authors comment if they had similar experience or how this could be explained?

In summary, I would like to congratulate the authors for achieving what could prove to be a major milestone.

Manuscript received September 7, 1991.

Fernando L. Alvarado and Claudio Cañizares (The University of Wisconsin, Madison, Wisconsin): This is an important paper on a very timely topic. These discussers want to add some additional comments to this topic. The use

of V-Q sensitivity factors is but one way of assessing proximity to collapse. A more direct way to assess this proximity is by simply measuring the distance to the point of system collapse directly in demand space, either along a correlated load increase direction or in the nearest direction to collapse. These discussers prefer this second approach, as it is more reliable and has better computational properties.

The paper is quite correct when it describes the necessity of correct system modelling: ordinary power flow models are not adequate. The proper models to use are the *dynamic* models for every component (including loads, regulators and controls), and the question to ask is whether the steady state equations resulting from this dynamic model are stable. Voltage instability is associated with a real eigenvalue becoming unstable.

The authors of this paper use a reduced jacobian matrix J_R , yet to perform the computation they (*correctly*) use the complete augmented set of equations to solve for the eigenvalues to prevent sparsity degradation. The reduction process does something different from what the authors imply in the paper. Elimination of the P equations not only reduces the problem, it also changes the nature of the assumed underlying dynamics. The implicit assumption is that the active power dynamics of the system play no significant role in voltage collapse. In this regard, the authors are probably right, and their reduction is indeed a good thing to do under most circumstances. However, there are some cases when one can produce a dynamic voltage collapse by "angle" bifurcations. A clarification of this point is needed.

Another problem with the proposed methodology is that the steady state equations used to determine the system state become singular at the bifurcation point. This produces numerical instabilities in the proposed method when close to the collapse point. Continuation methods present an alternative to resolve this problem, since they are numerically better behaved and yield an automatic way of increasing the load in the system [A4].

The authors demonstrate the usefulness of having eigenvector information, since at the point of collapse this vector pinpoints the area in which the collapse occurs. This has been shown in [A5] for a small system.

One problem with the author's methodology is that the eigenvalues closest to the axis, even when 10 of them are considered, are not necessarily the first ones that will eventually cross the axis. We are somewhat surprised that the authors have not looked more closely into the Point of Collapse (PoC) methodology. The PoC technique solves this problem by asking not what eigenvalue is closest to the real axis, but rather, at what value of a system load does any eigenvalue cross the real axis. In other words, it finds the most critical eigenvalue. The methodology was has been described in [A1] and further applications and implementations of this methodology have also been reported in [A2,A3]. The method is robust, reliable and has been implemented to solve systems of the same order as those described by the authors [A4]. Our experience has been that the PoC methodology is free from convergence difficulties and is reasonably fast even for very large systems.

[A1] F. L. Alvarado and T. H. Jung, "Direct Detection of Voltage Collapse Conditions," Proceedings of the 1988 International Conference on Bulk Power System Voltage Phenomena -- Voltage Stability and Security, Trout Lodge, Potosi, Missouri, September 18-24, 1988. Proceedings available as EPRI Report EPRI EL-6183, January 1989, pp. 5.23-5.38..

[A2] F. Alvarado, Y. Hu, D. Ray, R. Stevenson, E. Cashman "Engineering Foundations for the Determination of

Security Costs," IEEE Transactions on Power Systems, August 1991, Vol. 6, No. 3, pp. 1175-1182.

[A3] C. Cañizares, F. Alvarado, C. DeMarco, I. Dobson and W. Long, "Point of Collapse Methods Applied to AC/DC Power Systems," presented at the 1991 IEEE/PES Summer Meeting, San Diego, CA, July 29-August 1, 1991, paper 91SM 491-1-PWRS.

[A4] C. Cañizares and F. L. Alvarado, "Computational Experience with the Point of Collapse Method on Very Large AC/DC Systems," 1991 Deep Creek Lake Workshop on Voltage Stability and Security, August 5-7, 1991.

[A5] I. Dobson and H. D. Chiang, "Towards a Theory of Voltage Collapse in Electric Power Systems," Systems and Control Letters 13, 1989, pp. 253-262.

M. G. Lauby (Electric Power Research Institute, Palo Alto, Ca.): The authors are to be congratulated for a timely paper. It presents a novel approach to determine voltage stability of power systems. This discussor has a few comments and questions.

In the past, the words "voltage collapse" and/or "voltage instability" were employed to describe a power flow which would not solve after modeling a steady-state disturbance on the power system. However, as clearly demonstrated in your paper, one *can not* determine voltage stability with a common power flow. Rather, detailed models are needed to represent load, special controlled devices (i.e., FACTS Devices, HVdc, SVC, ULTC, etc.) and generation characteristics along with their shifts in reaction to system conditions. Indeed, even if a power flow does solve for steady-state conditions, voltage instability may exist for the modeled conditions as the path from the initial state to the final state can be voltage unstable.

By employing detailed steady-state models, and ignoring system dynamics, one can evaluate control coordination and its impact on voltage stability. Further, distance to voltage instability is also determined by the modal analysis. How/when would the authors suggest employing system stability analysis for assessing voltage stability, versus the modal analysis of the Jacobian? Namely, could the authors describe the voltage stability study process which employ both large signal stability and the pseudo-steady-state modal analysis methods?

Details describing the convergence characteristics of the snapshot power flow would be of interest. Also, what kind of power flow was employed?

Finally, power system engineers are not only interested in where and how far away the power system is from voltage stability, but also, how to remedy the problem. Can the modal analysis aid in directly determining design alternatives, rather than power system engineers "hunting" through a menu of system alternatives in an effort to eliminate voltage stability problems?

The authors are again complemented for an exciting paper. Their comments on the above is greatly appreciated.

Manuscript received September 3, 1991.

Miroslav Begovic (Georgia Institute of Technology, Atlanta), **Dejan Ostojic** (Belgrade University, Belgrade): The authors are to be commended for a well written and interesting paper. Similar approach, based on modal analysis of the Jacobian matrix was presented in [A1], where the power system was modeled as linearised system of equations in descriptor form, and eigen-analysis of the Q-V Jacobian submatrix was used to identify the critical modes, and, subsequently, the critical zones in the system, where remedial actions are needed. The authors of [A1] have used both sensitivity analysis of the eigenvalues of the Q-V Jacobian submatrix, and the sensitivity analysis of the generated reactive powers. The later approach is characterised by much smaller computational effort, since the sensitivity of generated reactive power Q_g (of a single machine, control area, or the whole system) with respect to parameter vector λ ($\lambda \in \mathbb{P} \subseteq \mathbb{R}^p$) is given by

$$\left(\frac{\partial Q_g}{\partial \lambda}\right) = \left(\frac{\partial Q_g}{\partial \lambda}\right) - \left(\frac{\partial Q_g}{\partial x}\right) \left(\frac{\partial F}{\partial x}\right)^{-1} \left(\frac{\partial F}{\partial \lambda}\right) = \left(\frac{\partial Q_g}{\partial \lambda}\right) - \left(\frac{\partial Q_g}{\partial x}\right) J^{-1} \left(\frac{\partial F}{\partial \lambda}\right)$$

where x ($x \in \mathbb{R}^n$) is the state vector, and $F(x, \lambda) = 0$ are the load flow equations. The Jacobian matrix J is not being inverted during calculation, because co-state x_c may be computed directly from

$$\left(\frac{\partial F}{\partial x}\right)^T x_c = \left(\frac{\partial Q_g}{\partial x}\right)$$

and the calculation reduced to

$$\left(\frac{dQ_g}{d\lambda}\right) = \left(\frac{\partial Q_g}{\partial \lambda}\right) - x_c \left(\frac{\partial F}{\partial \lambda}\right)$$

The sensitivity analysis of generated reactive powers can therefore be obtained with respect to bus injections, or branch flows, with very light computational effort [A2][A3], which allows its applications where time is a critical factor. We have obtained very similar results in trying to identify the critical zones with both modal analysis, and sensitivity analysis of the reactive powers. Have the authors considered the sensitivity analysis of reactive powers w.r.t. loading parameters as an alternative, or a companion technique to modal analysis?

- [A1] M. Begovic, D. Ostojic, "Determination of Critical Zones for Voltage Instability in Power Systems", Proc. 29th IEEE Conference on Decision and Control, Honolulu, Hawaii, December 1990, pp.3011-3013.
- [A2] M. Begovic, A. Phadke, "Control of Voltage Stability Using Sensitivity Analysis", paper 91 WM 231-1 PWRs, IEEE PES Winter Meeting, New York, February 1991
- [A3] M. Begovic, X.Y. Chao, A.P.S. Meliopoulos, "Voltage Stability Margin Enhancement Via Optimal Power Flow", Proc. NSF International Workshop "Bulk Power System Voltage Phenomena: Voltage Stability and Security", Wisp Resort, MD, August 1991 (to be published)

Manuscript received September 9, 1991.

B. Gao, G. K. Morison, P. Kundur: We wish to thank the discussers for their interest in our paper, and we very much appreciate their thoughtful comments. We will respond to each of the discussers separately.

Dr. Concordia: We agree that results obtained using a linearized approach, such as the modal analysis, must be interpreted with caution. For a comprehensive assessment of voltage stability, the system should be increasingly stressed, using powerflow analysis with enhanced modelling, until instability is reached. Modal analysis should then be conducted at selected operating points. Using this approach, the nonlinear nature of the system voltage stability characteristic is captured by the full powerflow solutions, and modal analysis provides information on areas prone to instability and critical system elements.

We also agree that a small positive eigenvalue (or large positive $V - Q$ sensitivity) is itself an indication of danger. It may be useful to specify voltage stability criteria in terms of both the absolute distance to instability and a lower limit on the minimum positive eigenvalue at a given operating point.

The modal analysis technique we have described in the paper is only one aspect of the voltage stability analysis package we have developed under the EPRI project RP3040-1. Other features of the overall package include facilities for automatically increasing load and generation to determine voltage stability limits, and computation of individual bus voltage stability indices for fast voltage stability screening.

For dynamic voltage stability analysis, we have enhanced the transient/midterm stability program (ETMSP) to include models for devices which are important for the study of system voltage stability, such as under load tap changers (ULTC), generator field current limiters and thermostatically controlled loads.

Mr. Mansour: We totally agree with Mr. Mansour that it is desirable for the program to be able to search for the non-trivial weak modes. One possible way to achieve such an educated search is to allow the user to specify the area of interest, in which case only the reactive

power variations at buses within the area of interest will be considered. We are currently planning to add this feature to the modal analysis program.

As for the divergence problem of the power flow program as the system approaches voltage stability critical point, our experience has always shown that the smallest eigenvalue, even though being positive in most cases, becomes very close to 0 (< 0.1 say) before power flow solution diverges. Important information regarding the dominant mode approaching voltage instability can always be extracted. If the system is stressed by increasing only the reactive demand, as is the case of creating a $Q - V$ curve for a single bus it is easier to obtain a power flow solution at unstable operating conditions.

Mr. Lauby: Compared with detailed time domain simulation, modal analysis has the advantage that it is computationally more efficient and it provides more insight into the mechanism of voltage instability. Modal analysis at different snapshots following major disturbances can be used to examine a wide range of system conditions and to provide guidelines regarding remedial actions. Large signal time domain simulation should be used for investigating specific voltage collapse situations, as well as coordination of protection and controls.

To obtain the snapshots discussed in the paper, a conventional power flow program was used, and divergence has not been a problem. To overcome the deficiency in device modelling in the conventional power flow, we have developed a power flow program with enhanced models for devices which are important for voltage stability studies. A better approach of obtaining snapshots of system conditions following major disturbances would be to solve for different quasi-steady states of the system based on the detailed formulation as used in time domain simulation program.

Bus, branch and generator participations provide important information regarding remedial actions to improve overall system voltage stability. For example, bus participations can help to determine siting of SVC installation and branch participations identify candidate lines for series compensations. This has been demonstrated in actual system studies as reported in Mr. Mansour's discussion.

Prof. Semlyen: The approach proposed attempts to determine the eigenspace corresponding to the minimum distance to voltage instability. We have found that the direction corresponding to the minimum distance to voltage instability most often represents unrealistic pattern of stressing the system and therefore is of little practical value. For planning studies, the way or "direction" in which load grows in actual system is well defined. Therefore the approach which is most practical, and used in the paper, is to determine the proximity to voltage instability by increasing load along this direction.

Although we have not had the opportunity of testing Professor Semlyen's computational procedure for studying large systems, it is not evident that the eigenspace approach offers better efficiency than the inverse iteration approach as proposed in the paper.

Prof. Alvarado and Mr. Cañizares: The elimination of part of the Jacobian matrix associated with the active power variation simply serves the purpose of reducing the degree of freedom for perturbing the system. Only reactive power variation is considered because (i) it is well known that reactive power generally has larger impact on bus voltage magnitude and (ii) system voltage control is almost exclusively achieved by managing reactive power. As pointed out in the paper, the effect of real power change on system voltage stability is taken into account by studying the incremental relationship between Q and V at different operating conditions. The program developed provides the option of modal analysis using the full Jacobian matrix. Our experience has shown that the singularity of the full Jacobian matrix and the singularity of the reduced Jacobian matrix always coincide. However, if the full Jacobian matrix is used due to the large volume of data generated it is difficult to discern the relationship between bus voltage and bus power demand.

The question regarding the convergence of the power flow solution is the same as the one raised by Mr. Mansour and same answer applies. Continuation method provides an useful alternative for automatically changing system condition and we are planning to further investigate its practical application.

We have done some limited work in testing the PoC methodology. Our major concern with the approach is its lack of robustness and

efficiency in dealing with various limits and control actions. Another shortcoming of the PoC method is that no intermediate information is available regarding the process of the system's approaching voltage instability. In studying practical voltage stability problems it is very important to be able to follow, as the system condition varies, how the system reaches its voltage stability critical point. At the critical point, it is usually necessary to look at more than one weak mode.

Dr. Pal: Some fundamental questions regarding the modal analysis technique are raised.

The first question is whether the singularity of Jacobian matrix has anything to do with voltage stability. We have clearly shown in the paper that voltage stability, determined using the $V-Q$ sensitivity criterion proposed, is closely related to the eigenvalues of the reduced Jacobian matrix. Under the reasonable assumptions stated in the paper, mathematical proof is rigorous. Results of modal analysis have also been verified by time domain simulation results [A]. When dealing with voltage stability it is important to keep in mind the practical aspects of the problem; whether the results we compute using a particular analytical technique reflect actual system problems and the solutions we come up with are meaningful. It is pointless to get caught up in a debate over arbitrarily chosen definitions.

We disagree with the discussor's comments regarding how induction motor model is included in forming the Jacobian matrix. Equation (23) does not represent a constant impedance load, this is particularly true if $\Delta T_L = 0$ when the exponent $m = 0$. If $m = 0$ the motor torque is independent of the motor speed, and therefore it is incorrect to conclude that $\Delta T_L = 0$ results in $\Delta S = 0$. It is important to realize

that the induction motor model looks at only the linearized relationship between motor reactive power and bus voltage. Any effect of large change in slip or voltage would require reinitialization of the motor variables and recalculation of the $Q-V$ characteristic. The snap-shot approach, as described in [A], addresses the problem of voltage stability evaluation at different stages following major disturbances.

The comments on the effect of ULTC operation on overall load characteristics are generally valid and, following the presentation of the paper, the program has been enhanced to take into account the effect of load characteristics including ULTC operation.

Prof. Begović and Mr. Ostojic: We consider the use of $Q-V$ submatrix inappropriate for the analysis of cases close to voltage instability because, as has been pointed out by Dr. concordia in his discussion, the coupling between $P-V$ and $Q-\theta$ becomes more important as the system approaches voltage instability.

We have used sensitivity analysis of bus voltage w.r.t. bus reactive power injection and found there exist major limitations, as discussed in the paper. We expect that the same limitations apply to the discussor's generator reactive power sensitivity w.r.t. loading parameter.

Reference

- [1] G. K. Morison, B. Gao, and P. Kundur, "Voltage Stability Analysis Using Static and Dynamic Approaches", to be presented at the 1992 IEEE PES Summer Meeting, Seattle.